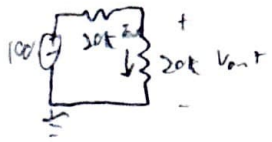
HW1

Dylan Thornburg



1a.

$V_{out} = 100\left(\frac{20k}{40k}\right) = 50V$

$I_{out} = \frac{100}{40k} = 2.5mA$

1b. Increase b/c of voltage division

$100\left(\frac{200k}{220k}\right) = 90.91$

90.91750

1c. No, it will get closer to 0, again b/c of voltage division.

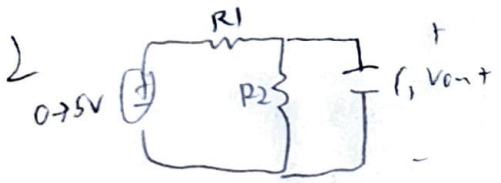
$100\left(\frac{20}{20020}\right) = .1V$

1d.

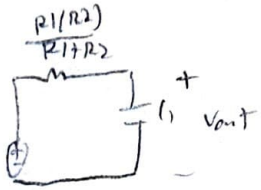
b would be the "on" or "pull up"

while c is the "off" or "pull-down"

2



$\rightarrow 5\left(\frac{R2}{R1+R2}\right)$



a. $R1 = R2 = 20k\Omega$, $C1 = .1mF$

Vout final = 2.5 V

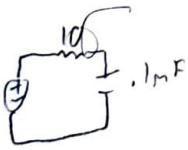
$\frac{2.5}{2} = 1.25$ $\quad 1.25 = 2.5\left(1 - e^{\frac{-t}{10000(.1\times10^{-6})}}\right) \rightarrow \frac{1}{2} = e^{\frac{-t}{10000(.1\times10^{-6})}}$

$\ln\left(\frac{1}{2}\right) = \frac{-t}{.001}$

$t = -.001\ln\left(\frac{1}{2}\right)$

$\boxed{t = 6.931\times10^{-4}s}$

b. $R1 = R2 = 20\Omega$, $C1 = .1mF$



$\frac{1}{2} = e^{\frac{-t}{10(.1\times10^{-6})}} \rightarrow t = -(1\times10^{-6})\ln\left(\frac{1}{2}\right)$

$\boxed{t = 6.931\times10^{-7}s}$

c. $R1 = R2 = 20k$, $C1 = 1mF$ $\rightarrow \frac{1}{2} = e^{\frac{-t}{10000(1\times10^{-6})}} \rightarrow t = -.01\ln\left(\frac{1}{2}\right) = \boxed{.006931s}$

d. They should be small (in this case 20).

e. The smaller a capacitor's capacitance, the quicker it will charge.

3. $V_{DSAT} = .6V$ $V_T = .43V$ $k_n = 115\ \mu A/V^2$ $\gamma = 0$ $\lambda = 0$ $V_{DD} = 2.5V$

$W = 10\ \mu m$ $L = .25\ \mu m$

$\frac{W}{L} = 40$

i. $V_{DS} = 1.25$, $V_{GS} = .625$

$1.25 > .625 - .43$ $\boxed{\text{saturation}}$ $\rightarrow I_D = \frac{1}{2} k' \frac{W}{L} (V_{GS} - V_T)^2$

$I_D = \frac{1}{2} 115 \times 10^{-6} (40)(.625 - .43)^2$

$\boxed{I_D = 8.746 \times 10^{-5} A}$

ii. $V_{DS} = .1V$ $V_{GS} = 1.25V$

$.1 < 1.25 - .43$ $\boxed{\text{linear region}}$ $I_D = k' \frac{W}{L} \left((V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right)$

$I_D = 115 \times 10^{-6} (40) \left((1.25 - .43)(.1) - \frac{.1^2}{2} \right)$

$\boxed{I_D = 3.542 \times 10^{-4} A}$

iii. $V_{DS} = 1.25$, $V_{GS} = .1V$

$.1 < .43$ so $\boxed{\text{cutoff} \quad I_D = 0A}$